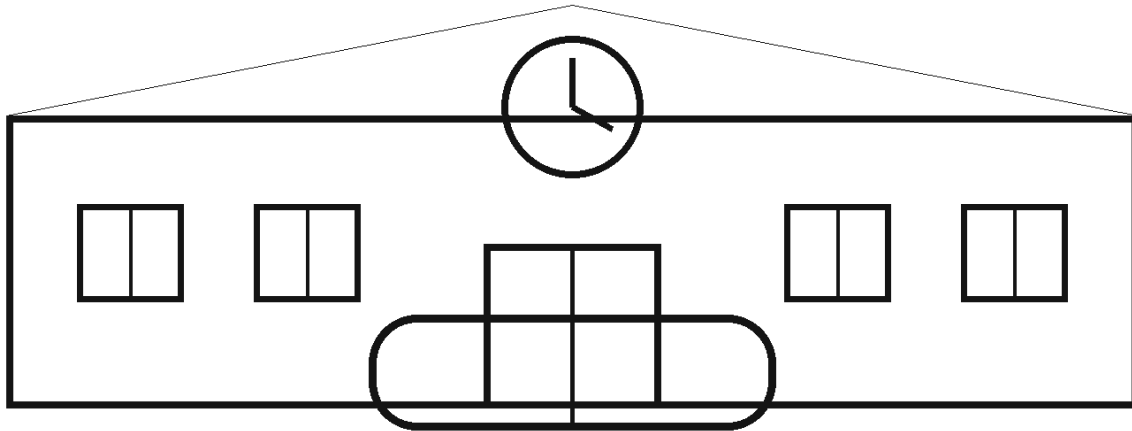


CASE FILE 01

THE MYSTERY OF THE MISSING TIME CAPSULE

A mixed-subject mystery for sharp-eyed Grade 6 investigators



YOUR INVESTIGATOR FILE

Name: _____ Division: _____ Date: _____ _____

☐ pencil ☐ eraser ☐ ruler ☐ colouring tools

MISSION

Follow the clues. Notice details. Check your math. At the end, explain what happened and prove it with evidence.

Do not jump to a conclusion. Good investigators change their minds when the evidence changes.

1 The Empty Display Case

TODAY'S JOB: Read the case. Mark facts and questions.

Monday, 8:35 a.m. At Cedar Ridge School in a fictional Surrey neighbourhood, the office discovers that the school time capsule is gone. It was sealed in a metal tube and placed in the courtyard display case on Friday. The case is locked. The glass is not broken. A storm passed over the school during the weekend.

The capsule holds student letters, photos, a school map, and small objects chosen to show life in 2026. It was supposed to be opened in 2046.

Your job is to answer three questions:

☐ Who moved it? ☐ Where is it now? ☐ Why was it moved?

FACT or QUESTION?

Statement	Fact	Question
The glass is unbroken.	☐	☐
Who has a key?	☐	☐
A storm happened.	☐	☐
Was the capsule damaged?	☐	☐

CASE BOARD

Draw the empty case. Add 3 details from the introduction.

CLUE A The first clue is a fact you think matters. _____

2

Witness Statements

TODAY'S JOB: Underline details. Infer carefully.

NOOR • library helper • 8:12 a.m.

“Near the courtyard I saw small wet wheel marks. They went toward the library hall. I did not see who made them.”

ELI • student • 8:20 a.m.

“I heard a cart rattle near the library. A blue sheet covered something long and round.”

MR. SINGH • custodian • 8:25 a.m.

“The courtyard roof leaked after the storm. I put out a warning sign. My tool cart stayed in storage.”

MS. ALVAREZ • librarian • 8:30 a.m.

“The archive-room door was open. I saw Ms. Chen returning an empty flat cart. She said, ‘The principal approved an emergency move.’”

SORT THE STATEMENTS

Directly observed	Someone reported it	Inference
wet wheel marks	emergency move	the cart carried capsule
☐ cart sound	☐ roof leaked	☐ move prevented damage
Add one: _____	Add one: _____	Add one: _____

An inference is a smart idea based on evidence. It is not yet a proven fact.

Best inference so far

I think _____ because _____.

PARTNER CHECKPOINT (optional): Compare one underlined detail. You may keep different ideas.

CLUE B Record the strongest witness detail. _____

3 Math Route

TODAY'S JOB: Calculate, follow the grid, and locate the cart.

Start at A1. Solve each step in order. Move only up, down, left, or right.

Example: $2.4 + 0.6 = 3.0$, so move 3 squares right.

Step	Calculate	Answer	Move
1	$1/2$ of 4	right	
2	$3.7 - 2.7$	down	
3	Perimeter of a square with 0.25 m sides	right	

Use this school grid. Put a dot at A1, draw the route, and circle the final square.

	1	2	3	4	5
A	Gym	Hall	Office	Stairs	Class
B	Garden	Courtyard	Library	Archive	Class
C	Field	Hall	Library	Storage	Music
D	Parking	Entrance	Hall	Workshop	Stage

PATTERN LOCK

The archive cart has tags: 3, 6, 12, 24, _____. Circle the rule.

☐ add 3 ☐ double ☐ add 6 ☐ multiply by 3

Next tag: _____ Challenge: What would the 7th tag be? _____

CLUE C Write the final grid square and room name. _____

4

Science Lab Notes

TODAY'S JOB: Use observations to suggest what the residue might be.

Investigators found clear drops and orange-brown flakes under the display case. Four safe tests were recorded.

Sample	Observation	Test result	Possible match
Clear drops	no smell	evaporates clean	water
Clear drops	pH 7	neutral	water
Brown flakes	rough, magnetic	sticks to magnet	iron-containing material
Brown flakes	does not dissolve	stays solid	iron-containing material

1. Circle the best conclusion:

☐ The case was near water. ☐ Paint spilled. ☐ The samples prove theft.

2. Evidence that supports it (choose two):

☐ pH 7 ☐ evaporates clean ☐ blue sheet ☐ cart sound

FAIR TEST PLANNER

Question: Which cover keeps a paper letter driest during the same gentle water spray?

CHANGE	type of cover
MEASURE	_____
KEEP THE SAME	amount of water, paper size, spray distance
PREDICTION	I think _____ will work best because _____.

Challenge: Why would changing the cover and the amount of water make the test unfair?

CLUE D Name the risk to the capsule. _____

5

Community History File

TODAY'S JOB: Read a map and timeline. Consider viewpoints.

Cedar Ridge School timeline (fictional)

1998	School opens. Room B4 is the records room.
2010	The records room becomes a locked archive.
2026 • Friday	Students seal the time capsule in the courtyard case.
2026 • Weekend	A windstorm damages part of the courtyard roof.
2026 • Monday	Principal approves an emergency safety move at 8:05 a.m.

The old map labels B4 “Records.” The new map labels B4 “Archive.” What changed? What stayed the same?

Show your thinking

Changed: _____ *Stayed the same:* _____

TWO PERSPECTIVES

Students	“We should have been told before it moved.”
Staff	“We had to act quickly to protect everyone’s work.”

Circle one fair next step: post an update / hide the move / blame a student

Why is your choice fair?

It is fair because _____.

CLUE E Record what the timeline confirms. _____

6 The Coded Note

TODAY'S JOB: Crack the message left beside the case.

Each coded letter is one step AFTER the real letter. Move each letter BACK one place.

Example: $D \rightarrow C$ $B \rightarrow A$ $A \rightarrow Z$

CODED MESSAGE

Q	S	P	U	F	D	U	G	S	P	N	X	B	U	F	S
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Write the three decoded words:

Message _____ _____ _____
--

HIDDEN WORD CHECK

Circle the first letter of each correct answer, then read down.

1. A neutral pH is... A • seven B • ten
2. The final math room is... R • Archive S • Gym
3. The flakes may contain... C • iron D • sugar
4. A smart idea from evidence is an... H • inference J • rumour
5. A fair test changes... I • one thing K • everything
6. A map key helps us... V • interpret symbols W • guess
7. Evidence should be... E • checked F • ignored

Hidden word: _ _ _ _ _

CLUE F Record the decoded message or hidden word. _____

7

Scene Reconstruction

TODAY'S JOB: Use visual details to show what likely happened.

Draw four comic panels. Use captions, arrows, symbols, or speech bubbles. Stick to the evidence; mark any imagined detail with a dotted outline.

1 • AFTER THE STORM	2 • THE RISK IS NOTICED
3 • THE EMERGENCY MOVE	4 • THE SAFE LOCATION

VISUAL CHECK

☐ cart or wheels ☐ water risk ☐ protective cover ☐ archive clue

Optional partner checkpoint: Point to one detail that came from evidence and one that you imagined.

CLUE G Write one visual detail that supports your solution. _____

8

ADST Design Challenge

TODAY'S JOB: Design a safer time-capsule display.

USER: students and staff **NEED:** protect the capsule while keeping it visible

CRITERIA (must do)

☐ keep water out ☐ allow viewing ☐ open for 2046

CONSTRAINTS (limits)

☐ fits one capsule ☐ uses common materials ☐ no electricity required

SKETCH • label at least 4 parts

Useful word bank: seal • hinge • latch • sloped roof • clear window • raised base • drain

Explain one improvement

My design protects the capsule by _____.

Challenge: Add a way to show if water entered without opening the case.

9

Logic Check

TODAY'S JOB: Use every clue. Eliminate choices that do not fit.

Four people were seen Monday morning. Only one person matches ALL three facts: had approval, used a flat cart, and returned from the archive.

Person	Had approval	Used flat cart	At archive	Fits all?
Mr. Singh	No	No	No	☐
Ms. Alvarez	No	No	Yes	☐
Ms. Chen	Yes	Yes	Yes	☐
Eli	No	No	No	☐

Circle the person who fits all three facts.

Reason

It must be _____ because _____.

RED HERRING CHECK

A red herring is a detail that distracts you. Which detail might wrongly make someone suspect Mr. Singh?

☐ He is the custodian. ☐ His tool cart stayed in storage. ☐ The map shows B4.

Why is a job title not proof? You may write or draw.

My thinking

A title is not proof because _____.

CLUE H Record who moved it. _____

10 Solve the Case

TODAY'S JOB: Explain what happened. Use evidence, not guesses.

EVIDENCE CHECKLIST • Review your clue strips A-H.

☐ witness detail ☐ math location ☐ science risk ☐ timeline fact ☐ decoded note ☐ logic result

WHO?	_____
WHERE?	_____
WHY?	_____
HOW?	_____

Write or draw your solution. Use at least 3 pieces of evidence.

CASE CLOSED

The time capsule was moved by _____ to _____ because _____.

Evidence sentence starters:

- The witness statement shows... • The math route led to... • The science notes suggest...
- The timeline confirms... • The decoded note says... • Together, these clues support...

DRAW THE FINAL SCENE

Include labels or a caption.

11 Early-Finisher Case Lab

TODAY'S JOB: Create something new. Choose ONE mission.

☐ A • Write a new witness statement ☐ B • Design a new code ☐ C • Create a time-capsule object

A • NEW WITNESS

Write a truthful statement containing one observation and one inference. Underline the observation; circle the inference.

Witness name: _____

"I noticed _____. I think _____ because _____."

B • NEW CODE

Invent a code for a 3-6 word message. Give one example so another student can solve it.

Code rule + message

Example: _____ *Coded message:* _____

C • CAPSULE OBJECT

Draw one object that could represent Grade 6 life today. Label it and explain why it belongs.

Object sketch and explanation

It belongs because _____.

Optional sharing: Trade only your new code or witness statement. Keep your final case solution with your own booklet.